

L2

CLASS X - SCIENCE

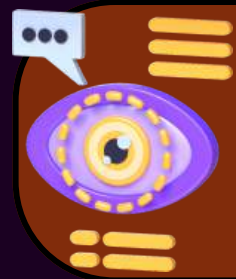
HUMAN EYE AND THE COLOURFUL WORLD

PRASHANT KIRAD



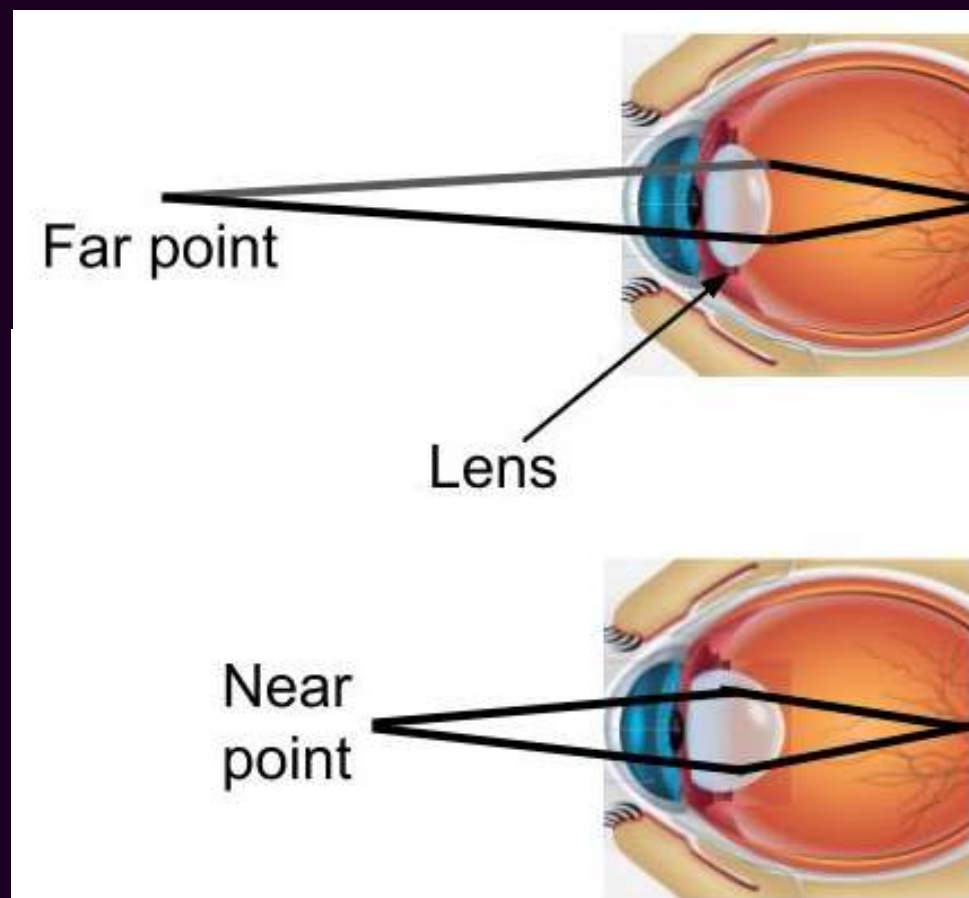
JOSH METER





Power of Accommodation

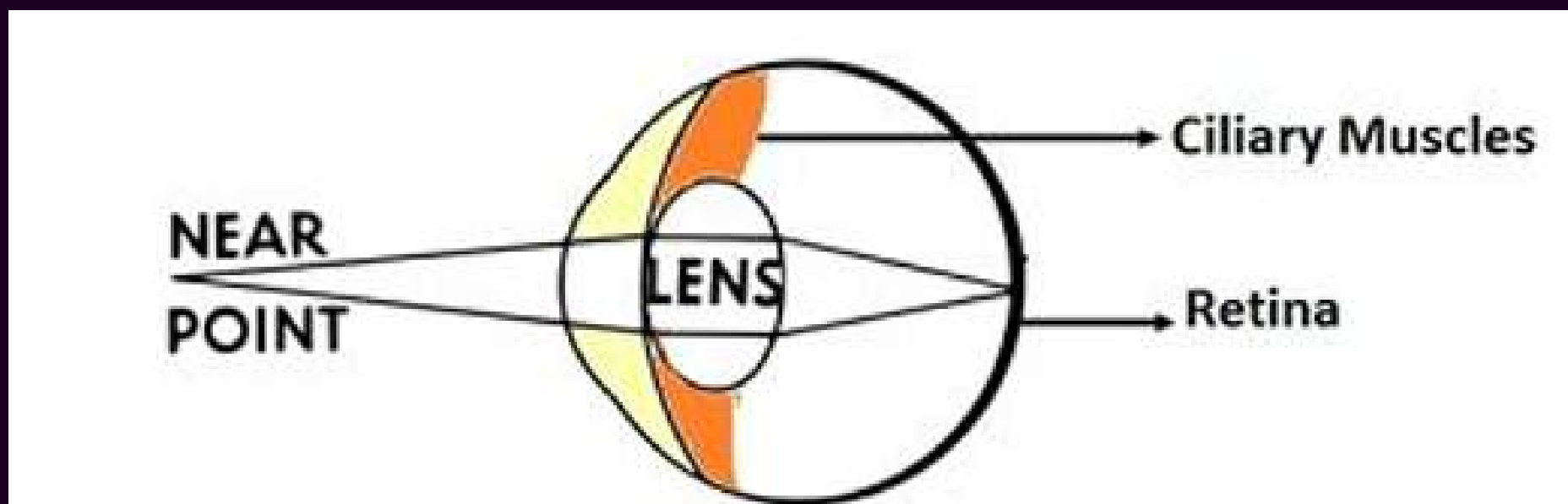
- The eye lens is flexible and its shape can be changed by the ciliary muscles.
- Changing the shape of the lens changes its **focal length**.
- The ability of the eye lens to adjust its focal length to clearly focus both near and distant objects on the retina is called the **power of accommodation** of the eye.



- Far Object
- Expanded Eye lens
- Focal length increases and power decreases
- **Ciliary muscles relax** → **lens becomes thin** → **focal length increases**
- Nearby Object
- Contracted Eye lens
- Focal length decreases and power increases
- **Ciliary muscles contract** → **lens becomes thicker** → **focal length decreases**

NEAR POINT

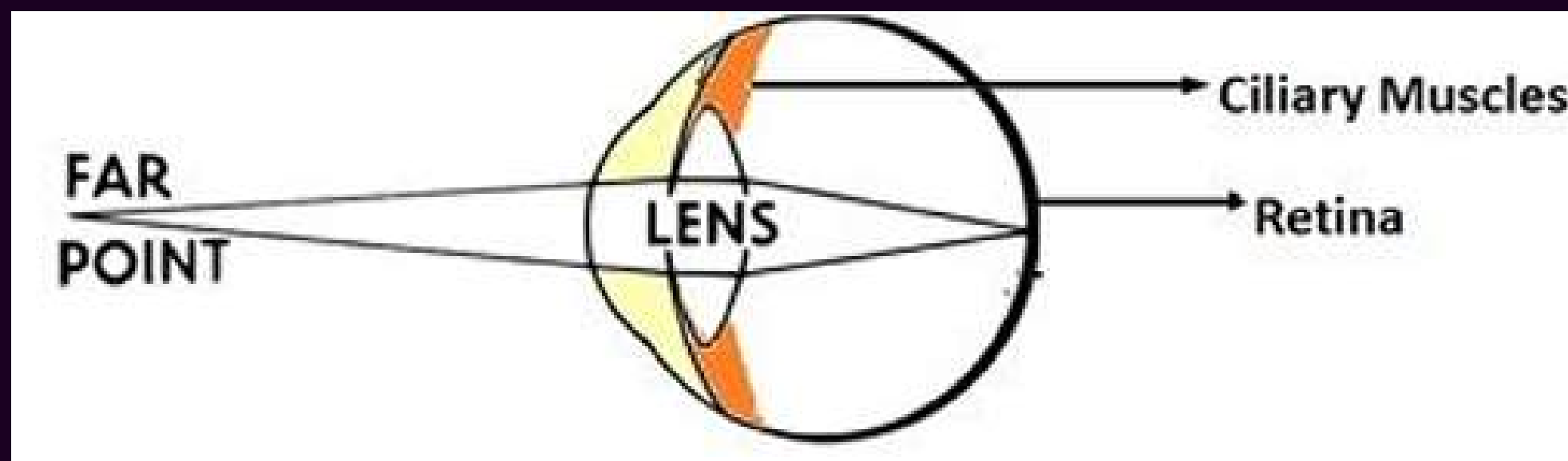
- **The minimum distance at which objects can be seen most distinctly without strain is called the near point of the eye.**
- **It is also called the least distance of distinct vision.**
- ***For a normal human eye, the near point is 25 cm.***
- **Below this distance, the image appears blurry and causes eye strain, as the lens cannot decrease its focal length further.**



Object Distance	Ciliary Muscles	Lens Shape	Focal Length	Result
Near Object	Contracted	Thicker	Decreases	Nearby object seen clearly

FAR POINT

- **The farthest point up to which the eye can see objects clearly is called the far point of the eye.**
- *For a normal eye, the far point is at infinity.*
- **The eye can see all objects clearly between the near point (25 cm) and far point (infinity).**



Object Distance	Ciliary Muscles	Lens Shape	Focal Length	Result
Distant Object	Relaxed	Thinner	Increases	Distant object seen clearly

Important Terms:

1. **Least distance of distinct vision (Near point):**
Minimum distance for clear vision without strain
= 25 cm for a normal eye.

2. **Far point:** Farthest point up to which the eye
can see clearly = Infinity for a normal eye.



Object Distance	Ciliary Muscles	Lens Shape	Focal Length	Result
Near Object	Contracted	Thicker	Decreases	Nearby object seen clearly
Far point	Relaxed	Thinner	Increases	Distant object seen clearly

Soch ke batao

Q.A person gets out in the sunlight from a dark room. How does his pupil regulate and control the light entering the eye?

- (a)** The size of the pupil will decrease, and less light will enter the eye
- (b)** The size of the pupil will decrease, and more light will enter the eye
- (c)** The size of the pupil will remain the same, but more light will enter the eye
- (d)** The size of the pupil will remain the same, but less light will enter the eye



Soch ke batao

Q. Define the term power of accommodation of human eye. Write the name of the part of eye which plays a major role in the process of accommodation and explain what happens when human eye focuses (i) nearby objects and (ii) distant objects. [CBSE 2024]

- The ability of the eye lens to adjust its focal length to clearly focus both near and distant objects on the retina is called the power of accommodation of the eye.
- Ciliary muscles play a major role in accommodation.

(i) Focusing nearby objects

- Ciliary muscles contract
- Eye lens becomes thick
- Focal length decreases

(ii) Focusing distant objects

- Ciliary muscles relax
- Eye lens becomes thin
- Focal length increases

Soch ke batao

- Q. (i) How does the change in curvature of the eye lens help us in the process of seeing the nearby objects clearly?**
(ii) State the range of the power of accommodation of a normal human eye.

- (i) While looking at nearby objects, the curvature of the eye lens increases. As a result, the focal length of the eye lens decreases, and the image is formed clearly on the retina.**
(ii) The range of vision of a normal human eye is from 25 cm near point to infinity far point.

Soch ke batao

Q. Define the term power of accommodation. Write the modification in the curvature of the eye which enables us to see the nearby objects. What are the limits of the accommodation power of a healthy normal human eye? [CBSE 2019]

- Power of accommodation is the ability of the eye lens to adjust its focal length to focus objects at different distances on the retina.
- To see nearby objects clearly, the ciliary muscles contract, making the eye lens thicker and more curved. As a result, the focal length decreases.

Limits of accommodation power of a normal eye

- Near point: 25 cm
- Far point: Infinity

Soch ke batao

Q. Why is a normal eye not able to see clearly the objects placed closer than 25 centimeter?

A normal eye cannot see objects clearly when they are placed closer than 25 cm because the ciliary muscles cannot contract further to increase the curvature and converging power of the eye lens. Therefore, the image cannot be focused clearly on the retina. This minimum distance of clear vision is called the near point of the eye.

Why Do Some Things Look Blurry?

Not everyone can see clearly at all distances.

WITHOUT GLASSES

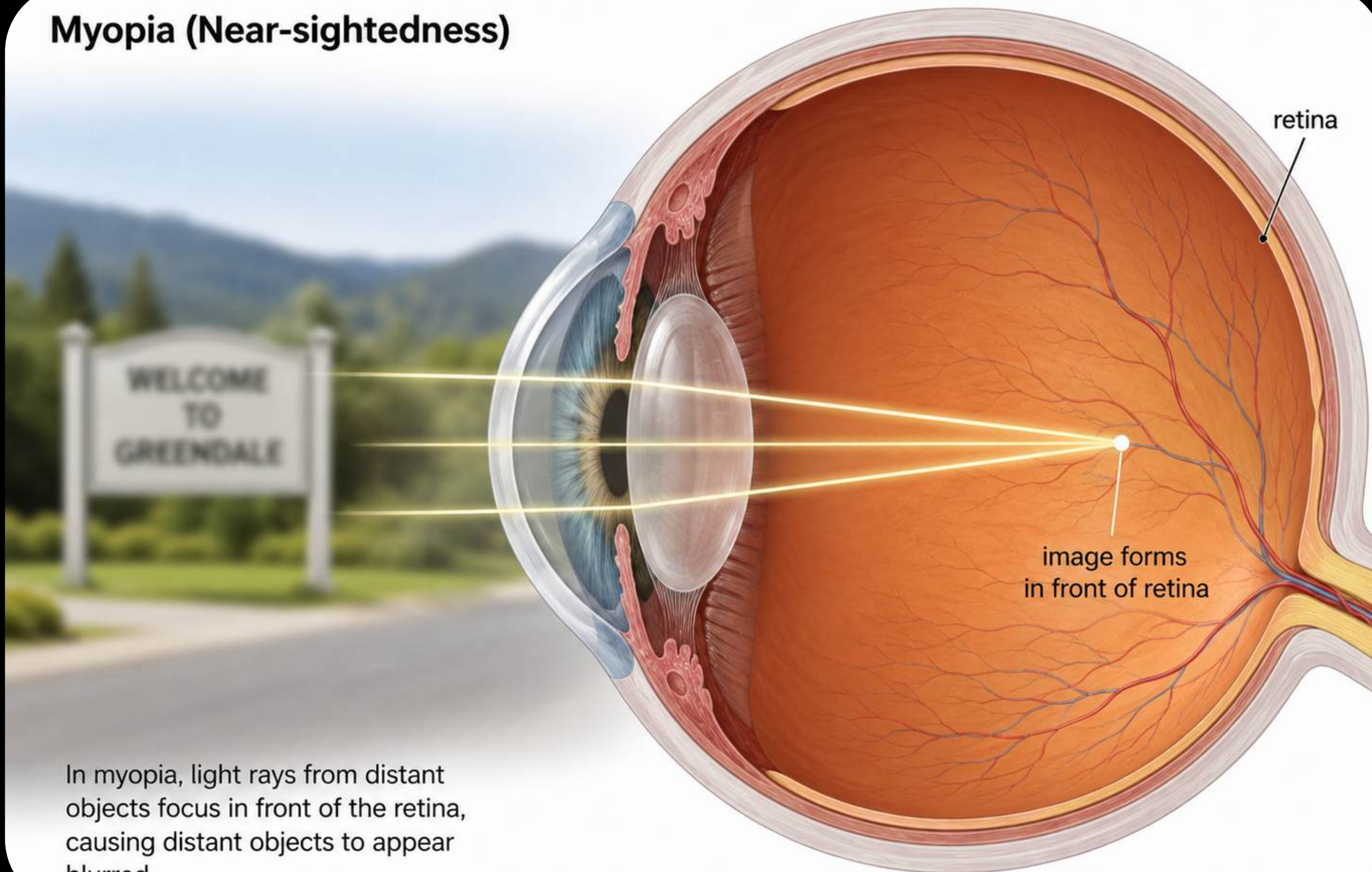


WITH GLASSES



Defects of Vision

Myopia (Near-sightedness)



In myopia, light rays from distant objects focus in front of the retina, causing distant objects to appear blurred.



Myopia (Near-sightedness)

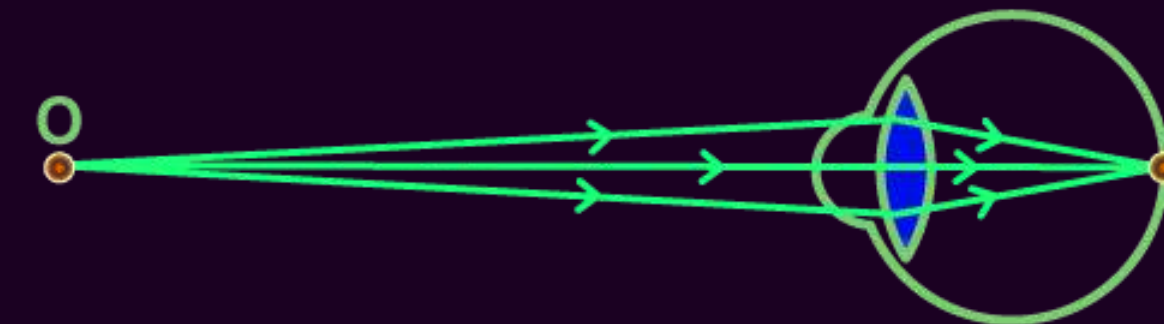
- Myopia is also called **near-sightedness**.
- A person **can see nearby objects clearly but distant objects appear blurred**.
- In a myopic eye, the image of a distant object forms in front of the retina.

Causes

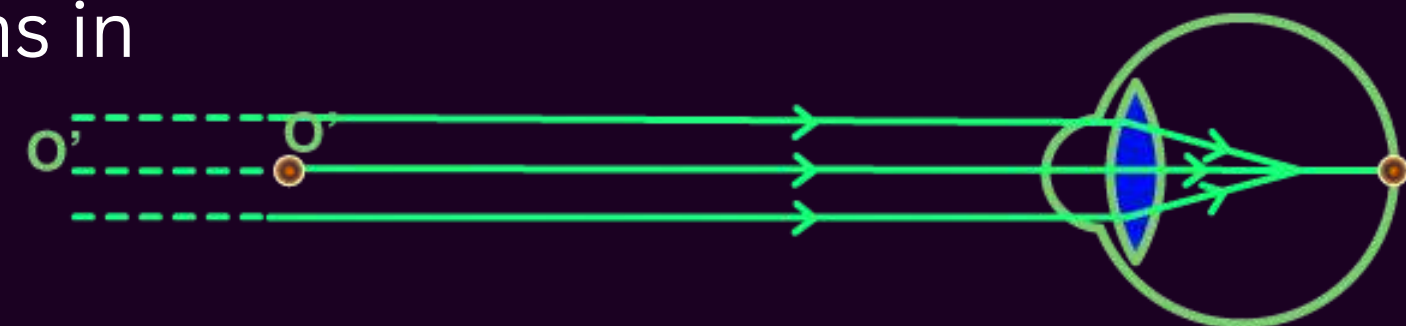
- excessive curvature of the eye lens, or
- elongation of the eyeball.

Correction

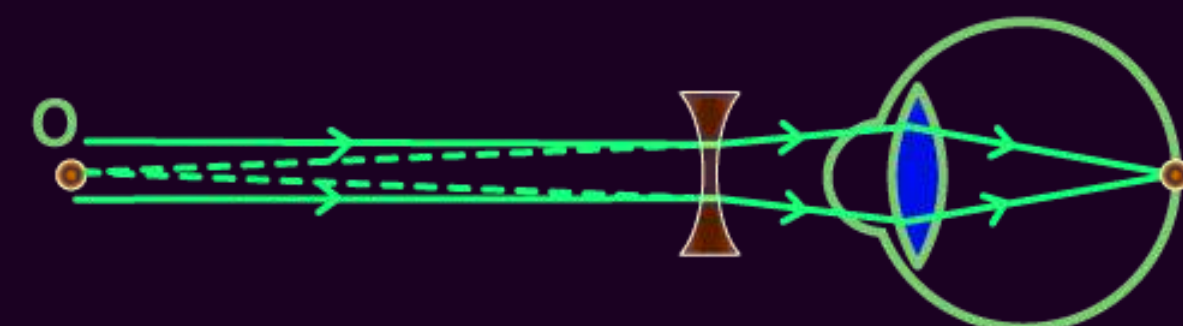
- Corrected using a concave lens.
- The concave lens helps focus the image on the retina.



(a) Far point of a myopic eye



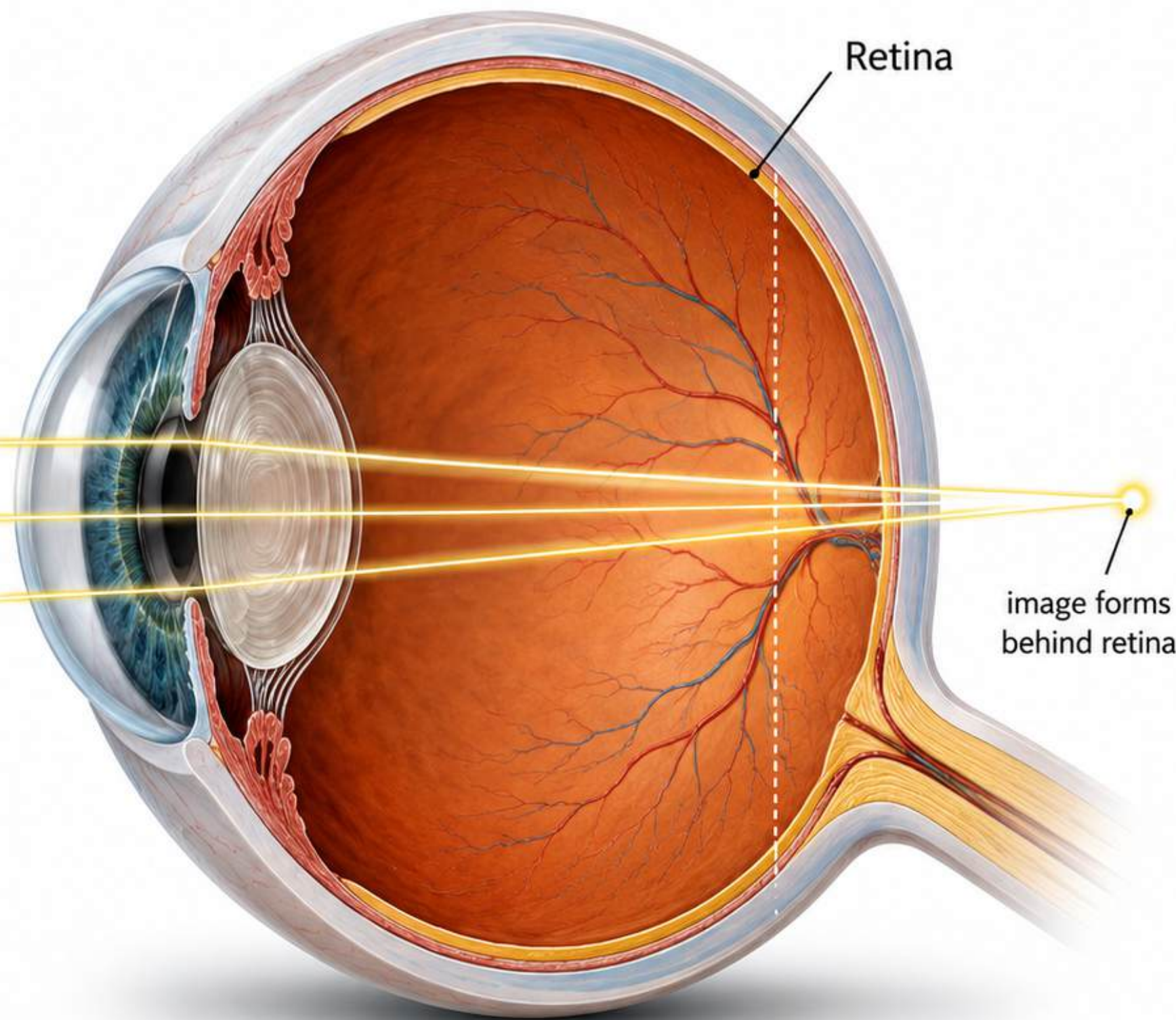
(b) Myopic Eye



(b) Correction for Myopic Eye

Hypermetropia (Far-sightedness)

NEAR OBJECT
(e.g., book or reading card)



In hypermetropia, light rays from nearby objects focus behind the retina, causing nearby objects to appear blurred.



Hypermetropia (Far-sightedness)

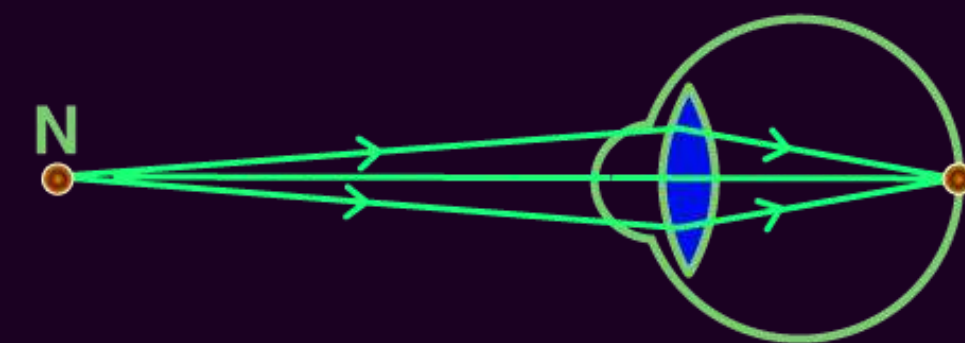
- Hypermetropia is also called **far-sightedness**
- A person can see distant objects clearly but nearby objects appear blurred
- The **near point** of the eye is farther than 25 cm
- In a hypermetropic eye, the image of a nearby object forms behind the retina

Causes

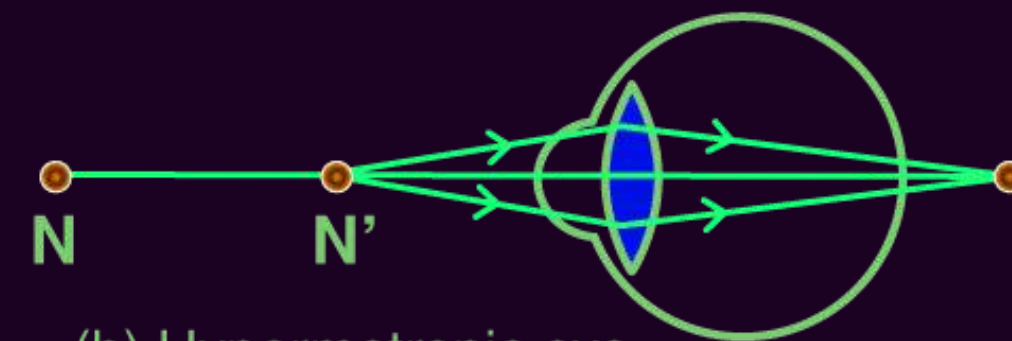
- Focal length of the eye lens becomes too long
- Eyeball becomes too small

Correction

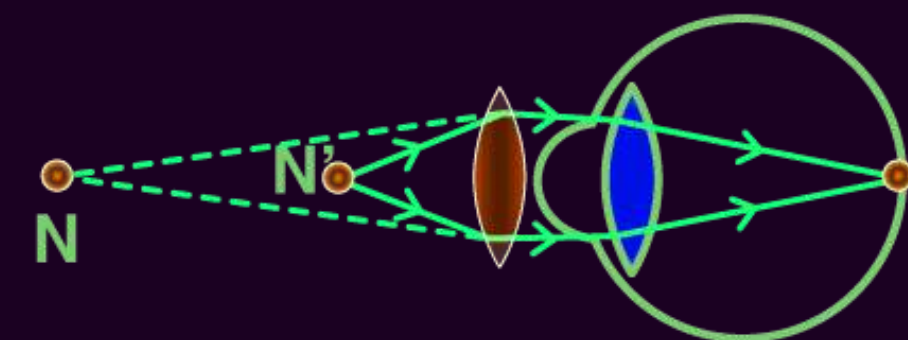
- Corrected using a convex lens.
- The convex lens helps focus the image on the retina.



(a) Near point of a Hypermetropic eye



(b) Hypermetropic eye



(b) Correction for Hypermetropic eye



Presbyopia (Age-related Far-sightedness)

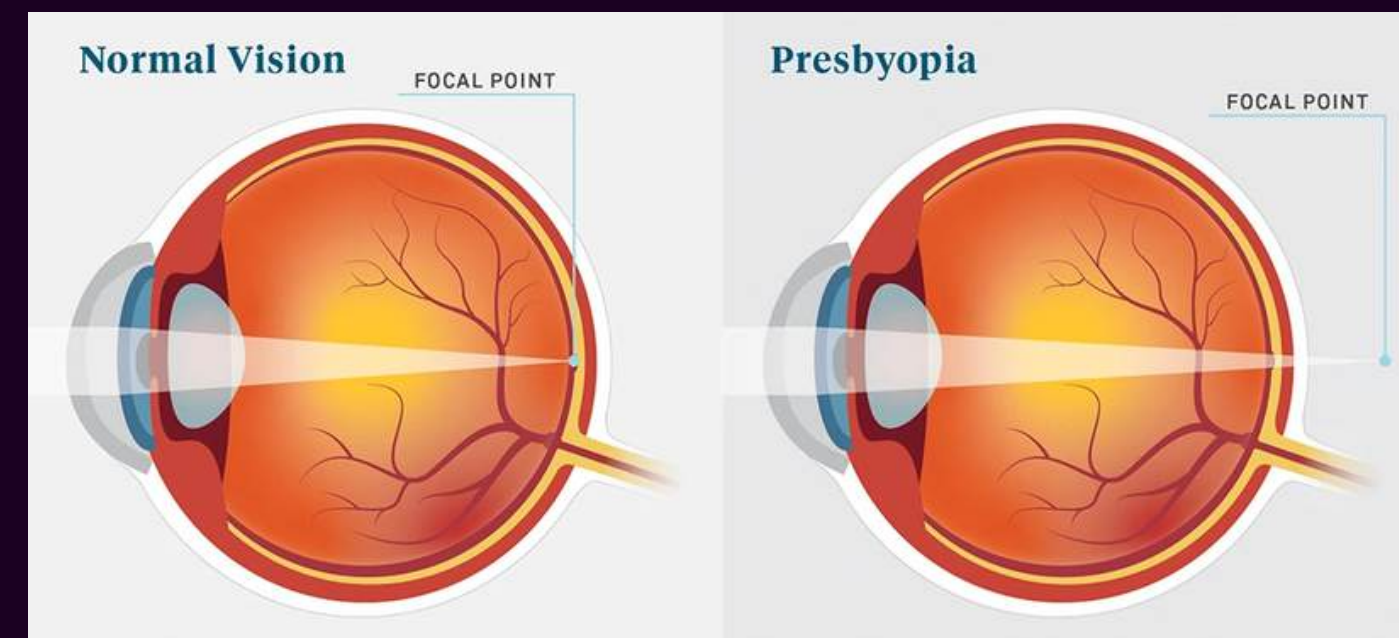
- Presbyopia is an eye defect that occurs due to ageing.
- In this defect, a person cannot see nearby objects clearly.
- The near point moves farther away from the eye.

Causes

- Weakening of ciliary muscles
- Loss of flexibility of the eye lens

Correction

- Corrected using bi-focal lenses.
- Upper part: Concave lens for distant vision
- Lower part: Convex lens for near vision





Astigmatism

- Astigmatism is a defect of vision in which **objects appear blurred or unclear.**
- It is caused by the irregular curvature of the cornea or eye lens.
- A person cannot see horizontal and vertical lines clearly at the same time.

Causes

- Uneven shape of the cornea or lens

Correction

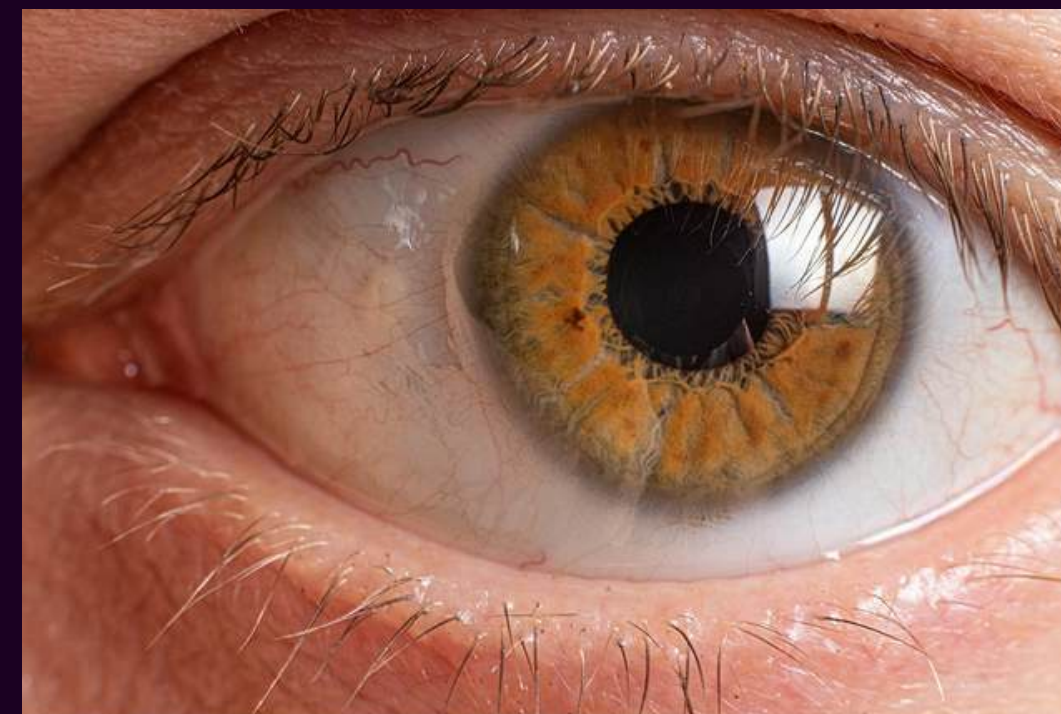
- Corrected using a cylindrical lens.



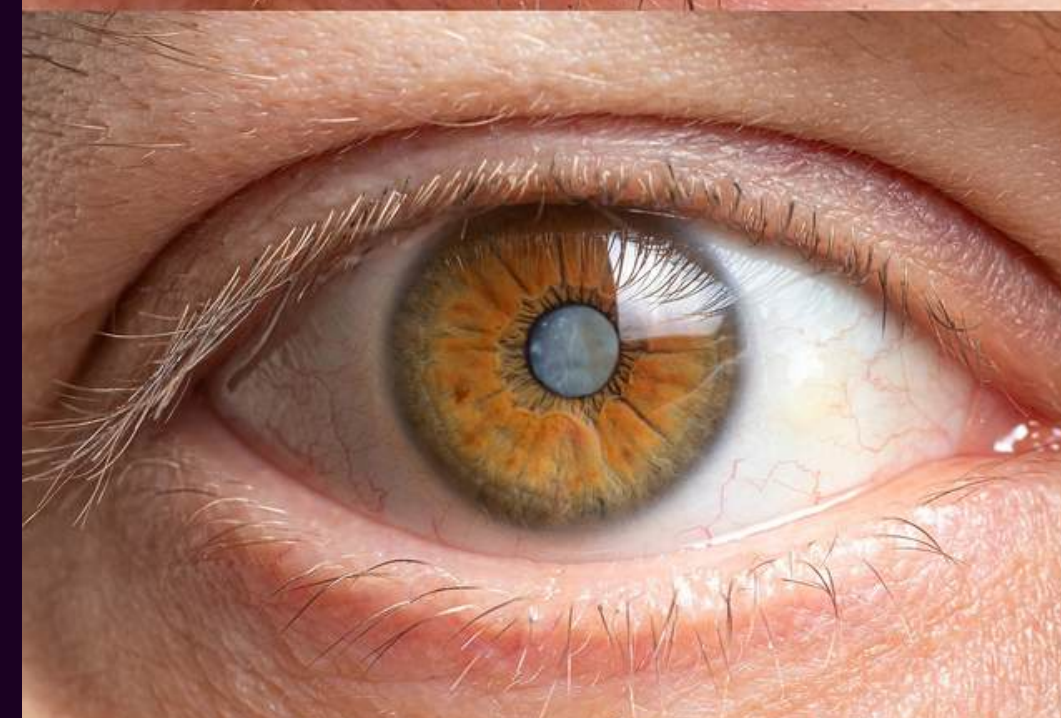


Cataract

- A condition in which crystalline lens of eye becomes milky and cloudy due to growth of membrane over it.
- It generally occurs in **older adults**.
- This causes partial or complete loss of vision.
- Vision can be restored through cataract surgery.



Normal
Eye



Cataract

Soch ke batao

Q. Assertion (A): Myopic eye cannot see distant objects distinctly.

Reason (R): For the correction of myopia, converging lenses of appropriate power are prescribed by eye surgeons. [CBSE 2024]

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

Soch ke batao

Q. Myopia can be caused due to: [CBSE 2014, 2016]

- a) Shortening of eye ball
- b) Weakening of retina
- c) Elongation of eye ball
- d) Thickening of cornea

Soch ke batao

Q. A person is unable to see distant objects clearly, but can read a book easily. What defect does he suffer from? [CBSE 2012, 2014, 2018]

- a) Hypermetropia
- b) Astigmatism
- c) Myopia
- d) Presbyopia

Soch ke batao

Q. Hypermetropia is caused due to: [2012, 2014]

- a) Elongation of eye ball
- b) Thick eye lens
- c) Shortening of eye ball
- d) Weak retina

Soch ke batao

Q. An old person is suffering from an eye defect caused by weakening of ciliary muscles and diminishing flexibility of the eye lens. If the defect of vision is 'a' which can be corrected by lens 'b', then 'a' and 'b' respectively are: [CBSE 2025]

- (a) Hypermetropia and convex lens**
- (b) Presbyopia and bifocal lens**
- (c) Myopia and concave lens**
- (d) Myopia and bifocal lens**

Soch ke batao

Q. Statement I: Presbyopia occurs due to weakening of ciliary muscles and reduced flexibility of the eye lens with age.

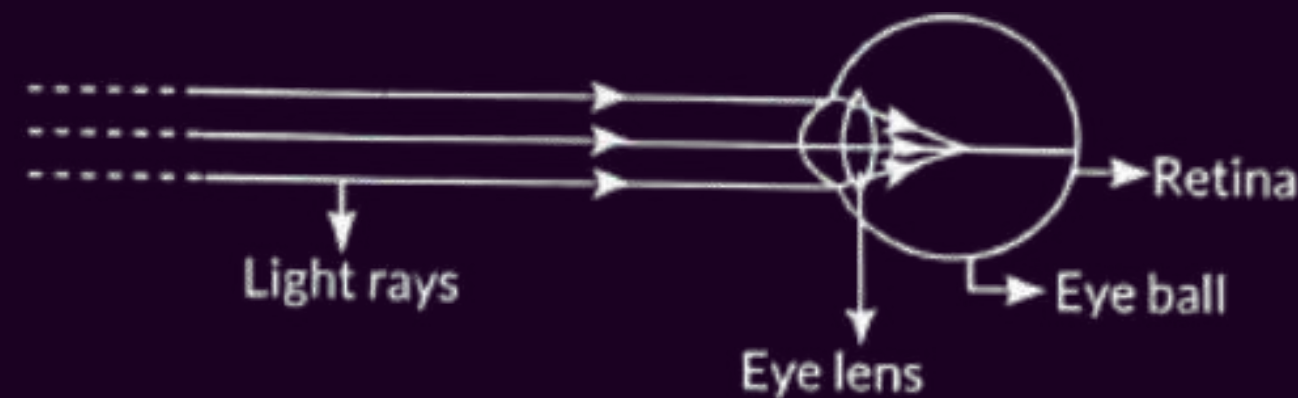
Statement II: Presbyopia is commonly corrected by using bifocal lenses.

Choose the correct option:

- (a) Both Statement I and Statement II are correct.**
- (b) Statement I is correct but Statement II is incorrect.**
- (c) Statement I is incorrect but Statement II is correct.**
- (d) Both Statement I and Statement II are incorrect.**

Soch ke batao

Q. Observe the following diagram and answer the questions following it: [CBSE 2019, 2023]



(i) Identify the defect of vision shown.

(ii) List its two causes.

(iii) Name the type of lens used for the correction of this defect.

The defect shown is Myopia (near-sightedness).

Reason: The image is formed in front of the retina. A person with myopia cannot see distant objects clearly.

(ii) **Two causes**

1. Excessive curvature of the eye lens
2. Elongation of the eyeball

(iii) **Correction:** Corrected using a concave lens.

Soch ke batao

- Q. A person cannot see the objects distinctly, when placed at a distance less than 50 cm.**
- (a) Identify the defect of vision**
 - (b) Give two reasons for this defect**
 - (c) Calculate the power of the lens he should be using to see the object placed at a distance of 25 cm from his eyes**
 - (d) Draw the ray diagram for the defective & corrected eye**
- [CBSE 2011]**

Soch ke batao

Q. (A) Rahul's doctor told him that the length of his eyeball is increased, i.e., the distance of retina from the eye lens has increased. Identify the defect of vision Rahul has, and suggest the correction of this defect with the help of a diagram.

(B) If 'x' is the distance of far point of defective eye, 'f' is the focal length of lens used for correcting this defect. Derive an expression for 'f'.

[DIKSHA]

Soch ke batao

Q. Varun instead of copying from the black board used to copy regularly from the notebook of his friend, Sudhir, with whom he sat on the same desk. Sudhir told the teacher about it. The teacher asked Varun to get his eyes checked by a doctor and explained to the whole class the reason why Varun copied from Sudhir's notebook.

(A) What, in your view, is wrong with Varun's eyes and how can it be corrected?

(B) If the doctor prescribes Varun to use lenses of power -0.5 D , what is the type of lenses?

(C) Write the values displayed by Sudhir and his teacher.

Soch ke batao

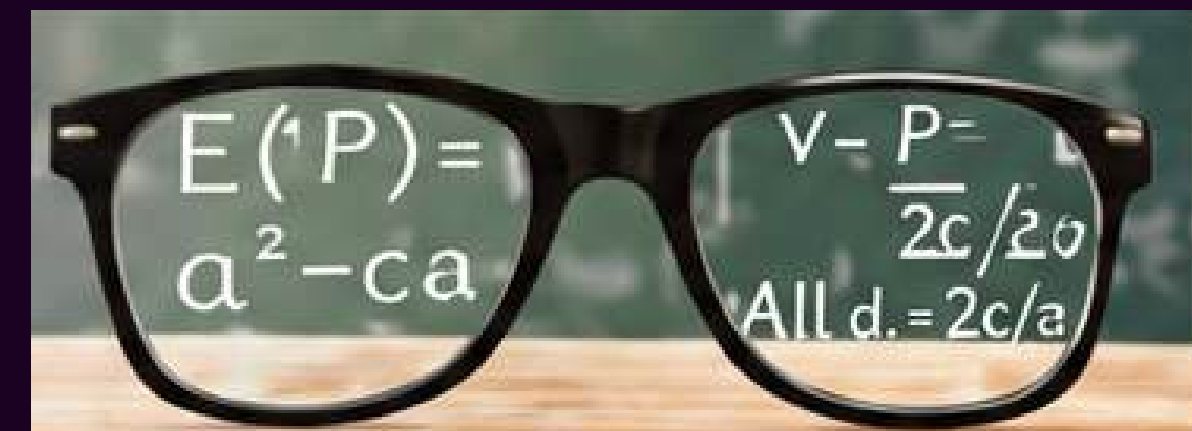
Q. Piyush, who was a back bencher in class, started complaining of frequent headaches. His parents took him to the nearest clinic and the doctor referred him to the eye specialist. The eye specialist tested his vision and asked Piyush whether he was able to read whatever the teacher wrote on the black board clearly or not. He replied in the negative. The doctor told his parents about the defect of vision that Piyush was suffering from and advised corrective glasses.

After wearing the glasses, Piyush was now able to read the blackboard clearly and also got rid of his headache.

(A) What type of defect is Piyush suffering from? What are the causes of this defect?

(B) Where is the image formed in Piyush's eyes? What type of lens is required to correct this defect?

Draw the (i) defected eye of Piyush (ii) correction for this defect.



Answer -

Answer

(A) Piyush is suffering from ****Myopia (short-sightedness or near-sightedness)**.

Causes of Myopia:

1. Elongation of the eyeball.
2. Excessive curvature (increased converging power) of the eye lens.

(B) In a myopic eye, the image of a distant object is formed in front of the retina.

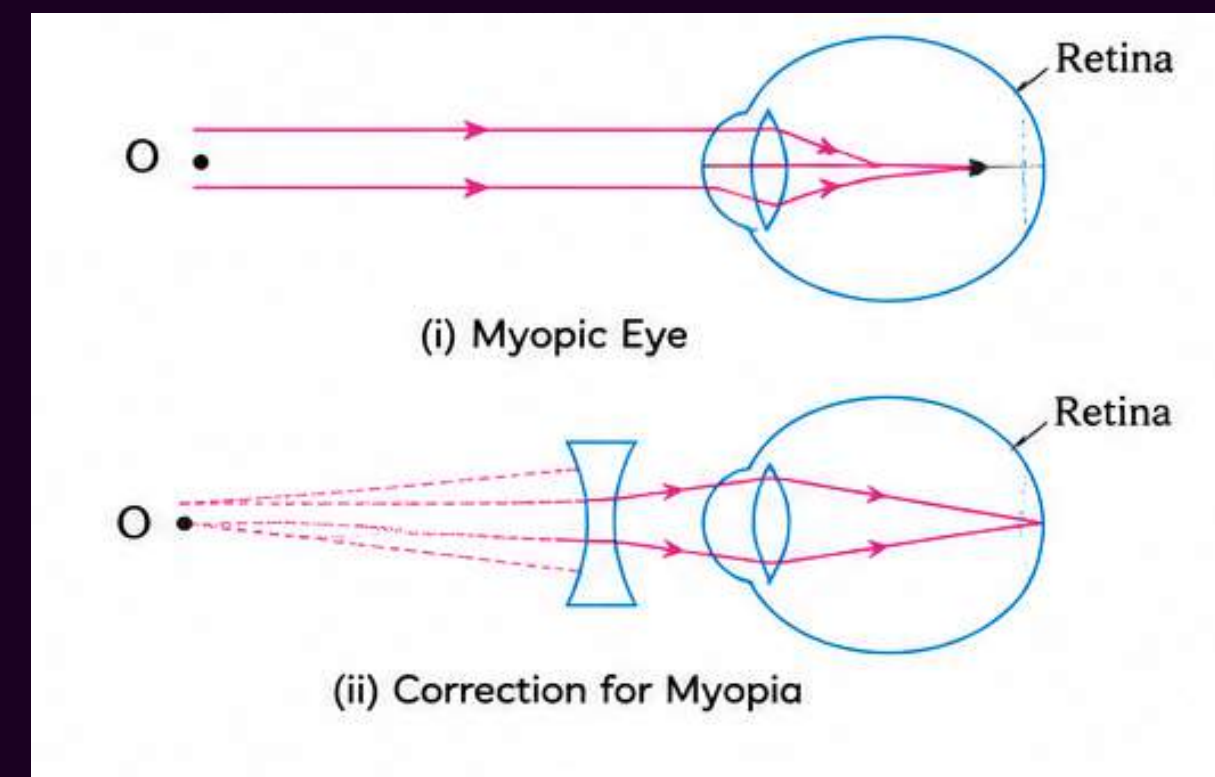
Correction:

The defect is corrected by using a ****concave (diverging) lens**** of suitable power. The concave lens diverges the incoming light rays so that, after refraction by the eye lens, the image is formed on the retina.

Ray Diagrams to draw

(i) Myopic Eye (image formed in front of the retina)

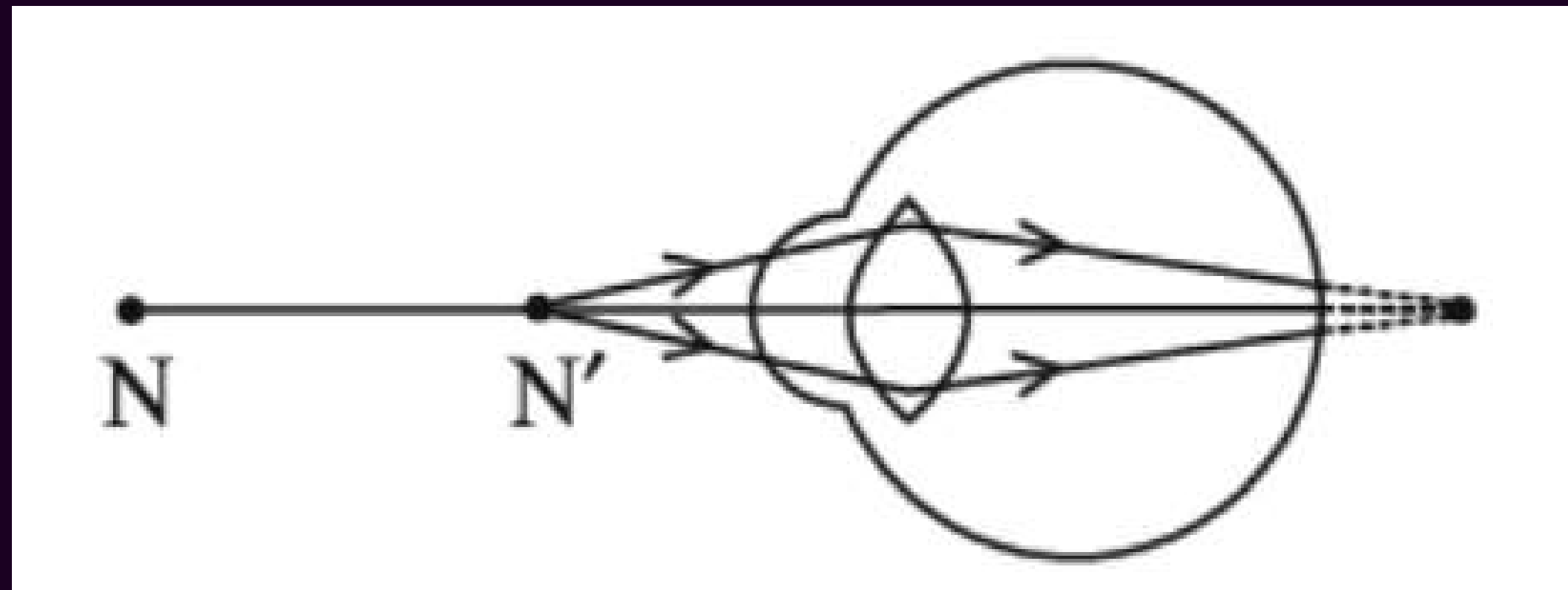
(ii) Correction of Myopia using a Concave Lens



Soch ke batao

Q. Study the diagram given below and answer the questions that follow: [CBSE 2024]

- (i) Name the defect of vision represented in the diagram. Give reason for your answer.
- (ii) List two causes of this defect.
- (iii) With the help of a diagram show how this defect of vision is corrected.



The defect shown in the diagram is Hypermetropia (far-sightedness).

Reason:

- The image is formed behind the retina.
- A person with hypermetropia can see distant objects clearly but cannot see nearby objects distinctly.

(ii) Two causes

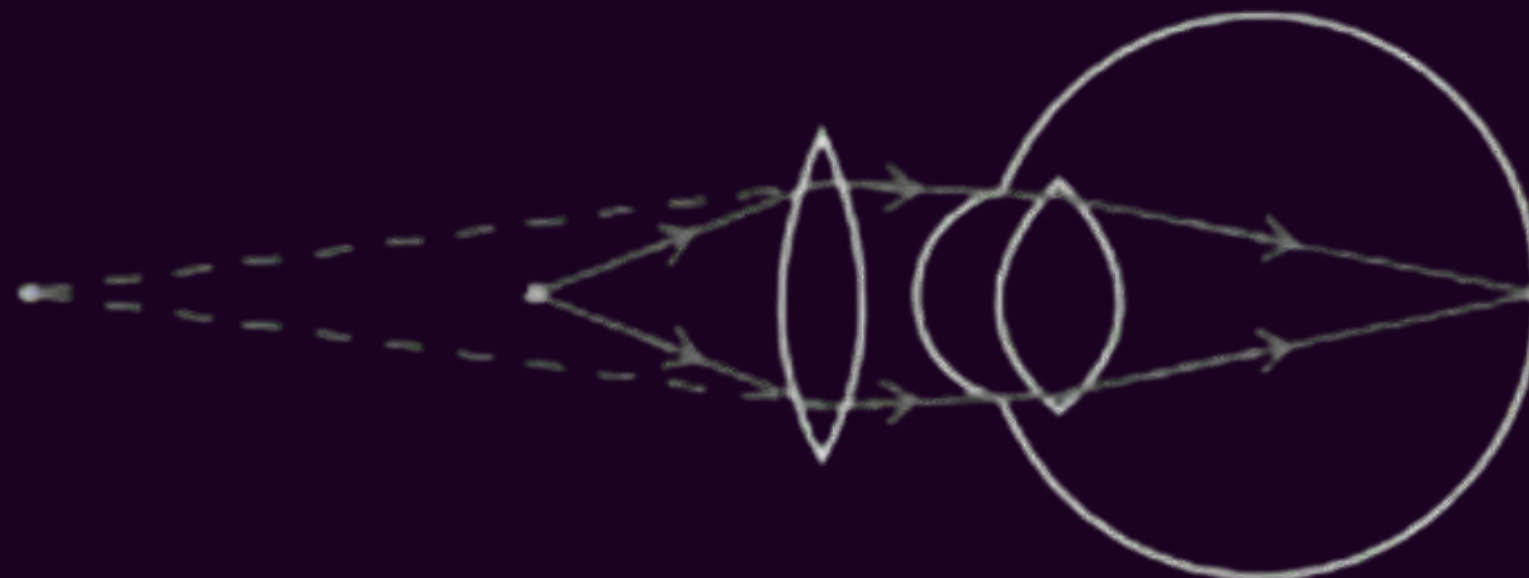
- Focal length of the eye lens becomes too long
- Eyeball becomes too small

(iii) Correction

- This defect is corrected using a convex lens.
- The convex lens helps to focus the image on the retina.

Soch ke batao

Q.



The above image shows a corrective measure for a particular defect of vision.

- (i) Identify the defect of vision and state what kind of lens is used to correct this deficiency.
- (ii) Draw and label a ray diagram that shows the defect of vision in the above case before correction.

Soch ke batao

Q. Name the type of defect of vision a person is suffering from, if he uses convex lenses in his spectacles for the correction of his vision. If the power of the lenses is +0.5 D, find the focal length of the lenses.

[CBSE 2017]

Soch ke batao

Q. (A) What is the fundamental difference between hypermetropia and myopia in terms of the optical experience of a person?

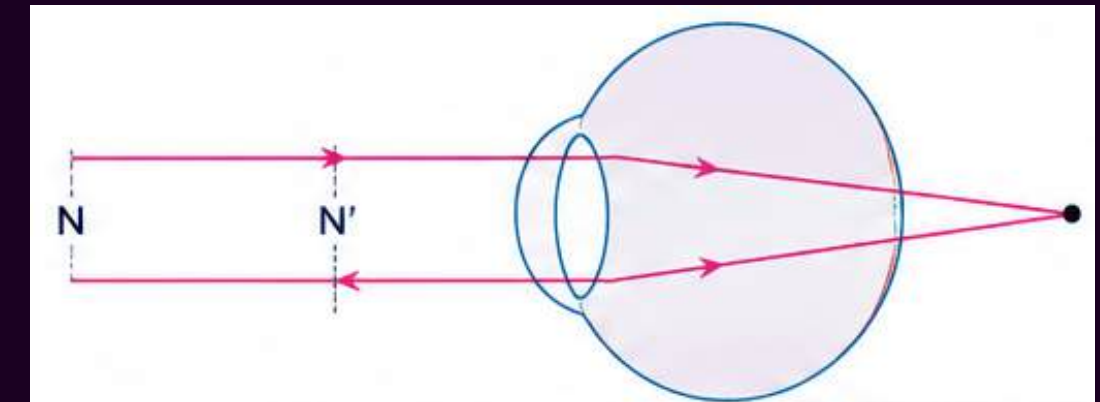
(B) The given diagram shows a special case of an eye defect.

(i) What is the defect that is shown in the figure?

(ii) State one cause for such a defect.

(iii) Explain with reason if a concave lens can be used to correct the defect.

[CBSE SQP 2024]



Soch ke batao

**Q. A person suffering from presbyopia needs spectacles having bifocal lenses. If the power of two lenses used in his spectacles is $+2.0\text{ D}$ and -0.5 D , which one of the two lenses is for the correction of his distant vision and what is its focal length?
[CBSE Compartment 2025]**

Soch ke batao

Q. (a) A person is suffering from both myopia and hypermetropia.

(i) What kind of lenses can correct this defect?

(ii) How are these lenses prepared?

(b) A person needs a lens of power $+3$ D for correcting his near vision and -3 D for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects.

[Outside Delhi Set I, 2020]

Soch ke batao

Q. Priya, a 58-year-old teacher, has trouble reading messages on her phone and also cannot clearly see the bus numbers from a distance. After an eye check-up, the doctor advised her to use bifocal lenses.

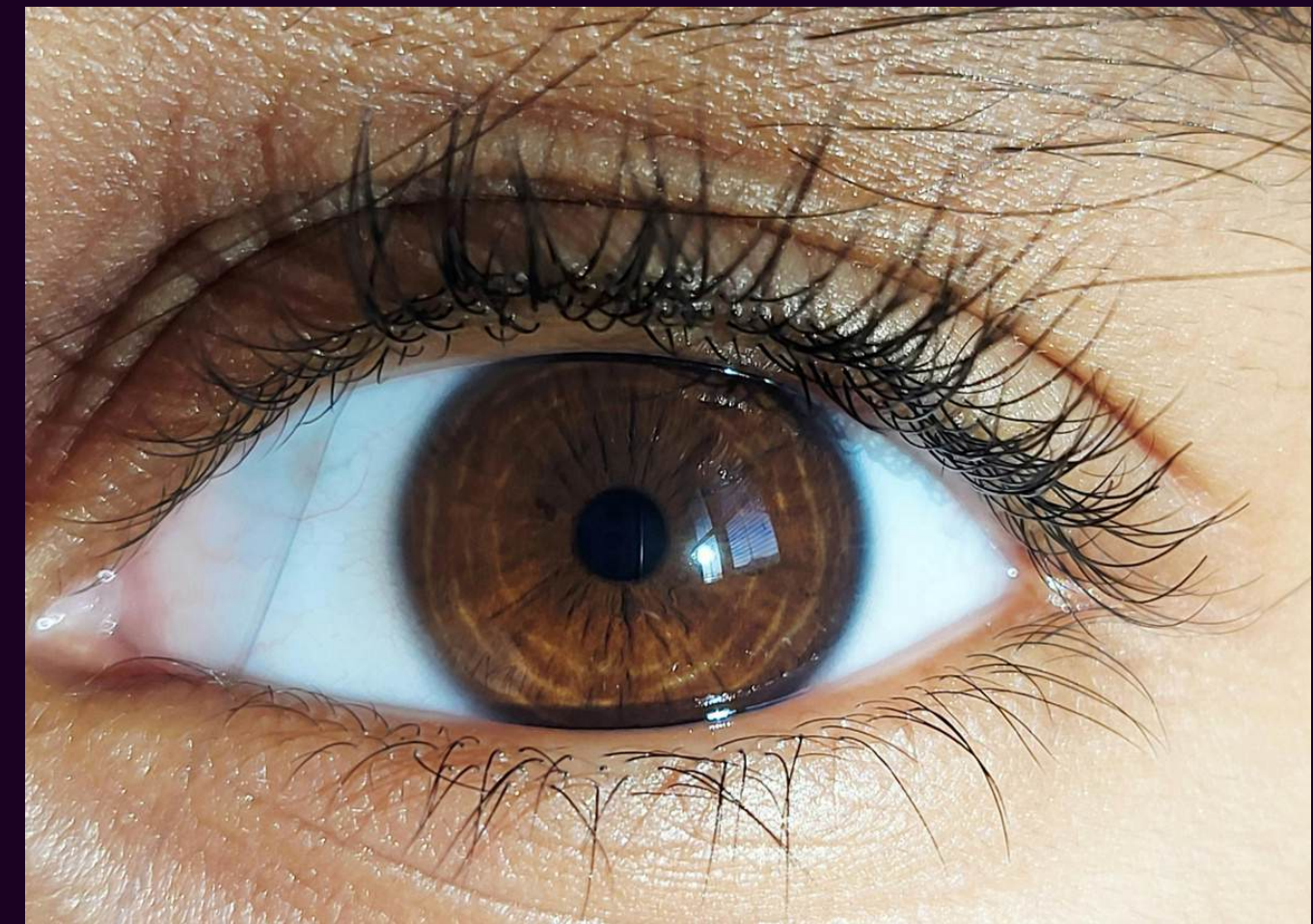
Questions:

- 1. Which defect of vision is Priya suffering from?**
- 2. Which type of lens is used to correct this defect?**
- 3. What are the causes of this defect?**

1. Priya is suffering from Presbyopia.
2. Bifocal lenses are used to correct this defect.
 - Upper part: Concave lens for distant vision
 - Lower part: Convex lens for near vision
3. Weakening of ciliary muscles and Loss of flexibility of the eye lens due to ageing

Think it over

- Our eyes can help someone even after our death.
- Donating eyes can give vision to corneal blind people.
- Eyes should be donated **within 4–6 hours after death**.
- People of any age or gender can donate eyes.
- Spectacle users, cataract-operated people, diabetics, BP and asthma patients can also donate eyes if they do not have communicable diseases.
- Eye removal takes only 10–15 minutes and does not disfigure the face.
- People who died due to diseases like AIDS, Hepatitis B/C, rabies, tetanus, cholera, meningitis, etc., cannot donate eyes.
- Eye banks collect, check and distribute donated eyes.
- One pair of eyes can give vision to up to four corneal blind people.



“You saw me and didn’t even say hi”

Me:

